K-Means Clustering Explained with an Example

K-Means is an **unsupervised machine learning algorithm** used to group similar data points into **K clusters**.

The algorithm tries to place data points into clusters such that:

- Points within the same cluster are similar.
- Points in different clusters are dissimilar.
- Each cluster is represented by its **centroid** (center point).

Example: Customer Segmentation

Suppose a retail store has customers with the following data:

Customer	Monthly Spending ($)
A	100
B	120
C	150
D	800
E	850
F	900

The store wants to identify customer groups.

Let's choose:

K = 2 (two clusters)

Step 1: Initialize Centroids

Randomly select two points as centroids:

- Centroid 1 = 120
- Centroid 2 = 850

Step 2: Assign Points to Nearest Centroid

Calculate distance from each point to both centroids.

Customer	Value	Distance to 120	Distance to 850	Cluster
A	100	20	750	C1
B	120	0	730	C1
C	150	30	700	C1
D	800	680	50	C2
E	850	730	0	C2
F	900	780	50	C2

Result:

Cluster 1: {100, 120, 150}

Cluster 2: {800, 850, 900}

Step 3: Recalculate Centroids

Compute the mean of each cluster.

Cluster 1

$$Centroid_1 = (100 + 120 + 150)/3$$

=123.33

Cluster 2

$$Centroid_2 = (800 + 850 + 900)/3$$

=850

Step 4: Reassign Points

Check distances again using new centroids.

Since all points remain in the same clusters, the algorithm has converged.

Final Clusters

Cluster	Customers
Cluster 1	A, B, C
Cluster 2	D, E, F

Visualization

```
100   120   150                                    800   850   900
 |-----Cluster 1-----|                              |-----Cluster 2-----|
       Centroid=123                                    Centroid=850
```

Real-World Applications

- Customer segmentation
- Recommendation systems
- Document clustering
- Image compression
- Market basket analysis
- Network traffic analysis
- Cell type identification in bioinformatics

How K-Means Works (Summary)

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1. Choose K clusters
2. Randomly initialize K centroids
3. Assign each point to nearest centroid
4. Recalculate centroids
5. Repeat Steps 3-4 until centroids stop changing
```

Mathematical Objective

K-Means minimizes the **Within-Cluster Sum of Squares (WCSS)**:

$$\sum_{i=1}^{K}\sum_{x \in C_i}||x-\mu_i||^2$$

Where:

- (K) = number of clusters
- (C_i) = cluster (i)
- (μ_i) = centroid of cluster (i)
- ($||x-\mu_i||^2$) = squared distance from point to centroid

The goal is to make points within a cluster as close as possible to their centroid.

Interview Question

How do you choose K?

A common method is the **Elbow Method**:

1. Run K-Means for different K values.
2. Calculate WCSS for each K.
3. Plot K vs WCSS.
4. Choose the point where the reduction in WCSS starts slowing down (the "elbow").

This provides a good balance between having too few and too many clusters.